RAJALAKSHMI ENGINEERING COLLEGE

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D2

Started on	Friday, 9 October 2026, 7:28 PM
State	Finished
Completed on	Friday, 9 October 2026, 7:30 PM
Time taken	1 min 37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n, a[1000], i;
    int slow, fast;

    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    slow = a[0];
    fast = a[0];

    do {
        slow = a[slow];
        fast = a[a[fast]];
    } while (slow != fast);

    slow = a[0];

    while (slow != fast) {
        slow = a[slow];
        fast = a[fast];
    }

    printf("%d\n", slow);

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔

	Input	Expected	Got	
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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